

report: BraTS 2026 Challenge task4

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1 Overview

1.1 LDM for inpainting

1.1.1 Latent Space Extension of Classical Inpainting Frameworks. The design of directly stacking noise and spatial conditions at the input level is inspired by two milestone works in conditional diffusion models: the **Palette** image-to-image diffusion framework [1] and the **RePaint** inpainting algorithm [2].

Literature has demonstrated that for generation tasks requiring strict **voxel-level spatial alignment**, such as lesion or artifact inpainting, early feature fusion via channel concatenation allows the first convolutional layer of the 3D U-Net to capture precise spatial correspondences immediately. This approach typically outperforms late fusion alternatives such as cross-attention mechanisms or auxiliary control networks.

To adapt this to the three-dimensional latent space, the input tensor of the primary denoising network in my study is configured as a 17-channel tensor:

$$\text{Input}_{\text{total}} = [\mathbf{z}_{\text{noisy}} (8 \text{ ch}) \parallel \mathbf{mask}_{\text{latent}} (1 \text{ ch}) \parallel \mathbf{z}_{\text{voided}} (8 \text{ ch})] \quad (1)$$

where $\mathbf{z}_{\text{noisy}}$ denotes the current latent state during denoising, $\mathbf{mask}_{\text{latent}}$ defines the spatial boundary for reconstruction, and $\mathbf{z}_{\text{voided}}$ represents the contextual patient anatomy outside the voided region.

1.2 Registration and Spatial Standardization

A critical prerequisite of the proposed pipeline is the integration of spatial standardization via image registration. operating generative models directly on heterogeneous native spaces often suffers from substantial structural variations across patients, leading to degraded inpainting quality. To mitigate this, this study enforces a bidirectional registration workflow utilizing the MNI152 standard template.

Before feeding the 3D MRIs into the Autoencoder, each subject's native volume is rigidly or non-rigidly aligned to the common MNI152 coordinate space. This preprocessing step ensures that identical anatomical regions across different patients are mapped to identical spatial coordinates within the tensor.

1.3 Anatomically-Guided Framework with Segmentation Feedback

To further enforce anatomical realism and prevent the hallucination of structurally implausible tissues within the masked regions, this study introduces a two-stage training paradigm inspired by the newly proposed Anatomically Guided Latent Diffusion Model (AG-LDM) [3]. The framework integrates a lightweight 3D tissue segmentation model, designated as WarpSeg, to provide explicit anatomical supervision.

1.3.1 Stage 1: Anatomically-Aware Autoencoder Regularization. In the first stage, rather than training a generic VAE/Autoencoder solely on voxel-level intensity reconstruction losses, we introduce a

semantic distillation mechanism. To implement this explicit anatomical supervision, a state-of-the-art 3D brain tissue segmentation model is retrieved from the **MONAI Bundle** repository. During the entire autoencoder fine-tuning process, this segmentation network serves as a **frozen teacher** model; its parameters θ_{teacher} are locked to prevent gradient updates, ensuring a stable and reliable anatomical reference.

1.3.2 Stage 2: Segmenter-Assisted Diffusion Trajectory Tuning. In the second stage, the latent diffusion model (LDM) is trained utilizing the established 17-channel input concatenation strategy ($\mathbf{z}_{\text{noisy}} \parallel \mathbf{mask}_{\text{latent}} \parallel \mathbf{z}_{\text{voided}}$) directly mirroring AG-LDM's philosophy of avoiding auxiliary control networks through input-level feature fusion. Crucially, to maintain morphometric integrity during the reverse denoising process, a segmenter-assisted sampling guidance is integrated.

During training, at every fixed interval of f_{seg} iterations, a short-trajectory Denoising Diffusion Implicit Model (DDIM) sampling sequence is executed to reconstruct intermediate latent states into the image domain. These intermediate samples are then evaluated by the frozen WarpSeg teacher. The resulting gradients from the segmentation consistency loss are propagated back to dynamically adjust the diffusion trajectory. This collaborative feedback loop acts as an anatomical anchor, ensuring that the 17-channel conditioned U-Net remains bounded by strict structural constraints throughout the entire generative reverse process.

2 Methodology

The proposed MNI-Based BraTS Inpainting Pipeline consists of a bidirectional spatial normalization loop and a two-stage anatomically-grounded generative framework.

2.1 Stage 0: Data Preprocessing and Unified MNI Cache Construction

Operating generative models directly on heterogeneous native clinical spaces suffers from substantial structural and geometric variations across patients. To eliminate global orientation variations and scale discrepancies, we develop a streamlined preprocessing engine via `build_mni_dataset.py`.

The data pipe standardizes raw structural BraTS volumes by estimating rigid/non-rigid transformation matrices that map each subject to the common MNI152 standard 1mm template.

2.2 Stage 1: Anatomy-Aware Autoencoder Regularization

The Stage 1 architecture constructs a high-fidelity latent space using a customized `AnatomyAwareVaeGanTrainer` backed by a PatchGAN discriminator. Rather than relying solely on raw intensity reconstruction constraints, a pre-trained 3D brain segmentation model from the **MONAI Bundle** ecosystem is introduced as a frozen teacher to regularize the training trajectory.

To enforce strict structural boundaries on tissue types (such as white matter and gray matter), the total loss incorporates a sparse

Training and Inference Pipeline

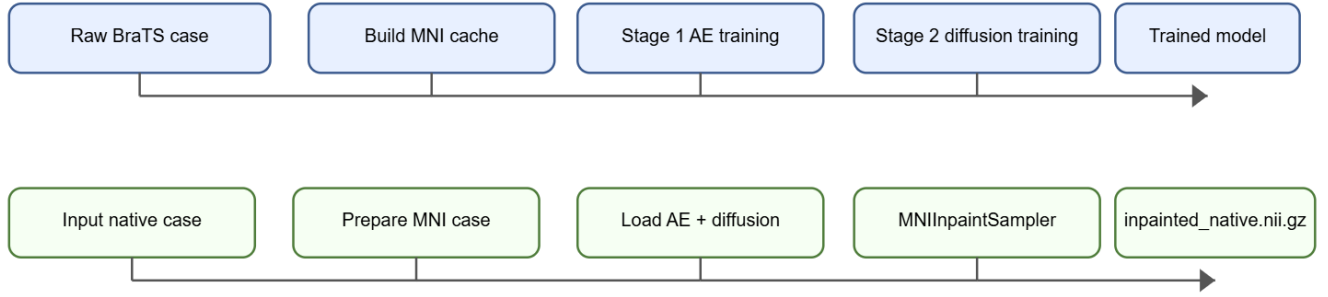


Figure 1: Pipeline

anatomical segmentation penalty calculated every N_{steps} :

$$\mathcal{L}_{\text{AE}} = \mathcal{L}_{\text{base}} + \lambda_{\text{seg}} \mathcal{L}_{\text{Seg}}(S(\hat{x}; \theta_{\text{teacher}}), S(x; \theta_{\text{teacher}})) \quad (2)$$

2.2.1 Low-VRAM Operational Constraints. Due to the massive memory footprint of 3D convolutional architectures, training is strictly constrained to prevent Out-Of-Memory (OOM) failures on standard 16 GB commercial GPUs. We employ an active low-VRAM mitigation strategy: setting the training batch size to 1, downscaling the sub-volume patch size to $[96 \times 128 \times 96]$, setting $\lambda_{\text{seg}} = 0.01$, and enforcing sparse execution via $N_{\text{steps}} = 4$ (seg_every_n_steps).

2.3 Stage 2: Mask-Conditioned Latent Diffusion Training

The Stage 2 architecture utilizes a MaskConditionedLDMTrainer operating under a strong latent conditioning paradigm. To preserve absolute patient-specific fidelity and avoid textural blurriness introduced by population-averaged templates, our configuration strictly bypasses external atlas priors (concat_atlas_latent=false) and relies entirely on local context conditioning.

The denoising network (3D U-Net) accepts a tightly-aligned 17-channel input tensor constructed by early-stage voxel-level concatenation:

$$\text{Input}_{\text{diffusion}} = [\mathbf{z}_{\text{noisy}} (8 \text{ ch}) \parallel \mathbf{mask}_{\text{latent}} (1 \text{ ch}) \parallel \mathbf{z}_{\text{voided}} (8 \text{ ch})] \quad (3)$$

Furthermore, to maintain morphometric integrity within the generated area during reverse sampling, a segmenter-assisted feedback loop is triggered every $f_{\text{seg}} = 20$ iterations. This loop executes a short-trajectory 10-step DDIM rollout, evaluates the intermediate structural boundary consistency via the frozen teacher, and back-propagates guidance gradients ($\gamma = 0.02$) to dynamically tune the diffusion trajectory.

2.4 Inference and Inverse Spatial Mapping

During the generative inference phase, the MNIImpaintSampler takes the standardized MNI test volume, generates the masked latent representations, and runs a full DDIM reverse trajectory with mask-preserving updates at each step.

The generated latent is mapped back to the image domain via the Stage 1 Decoder. To maximize tissue authenticity, a pixel-level composition boundary is applied: voxels outside the spatial mask are drawn directly from the original healthy background ($\mathbf{x}_{\text{voided_MNI}}$), while voxels inside the mask boundary are populated by the generated pseudo-healthy structures. Finally, the synthesized MNI volume ($\mathbf{y}_{\text{MNI}}$) is transformed using the inverse transformation matrix ($\mathcal{T}_{\text{mri} \rightarrow \text{mni}}^{-1}$), warping the final clean output seamlessly back into the patient’s original, native clinical space ($\mathbf{y}_{\text{native}}$).

2.5 Region-Aware Latent Diffusion Models

To address the topological variability of brain lesions, we partition the inpainting task into three localized sub-problems based on anatomical regions:

- **Deep Model:** Tailored for subcortical and deep gray matter structures, capturing localized interior context.
- **Cortical Model:** Optimized for superficial lesion masks, prioritizing fine-grained cortical surface features and sulcal topology.
- **Center Model:** Constrained to the ventricular and midline axis, enforcing global spatial symmetry.

2.6 Anatomically-Guided Mask Augmentation

To prevent overfitting to explicit training masks, we employ a region-constrained mask augmentation pipeline. Synthetic geometric primitives (spheres, boxes, and 3D blobs) are stochastically generated and injected exclusively into the target anatomical region corresponding to the active model. Random affine transformations and elastic deformations are applied before downsampling the mask to match the spatial resolution of the latent space $\mathbf{z}$.

3 Results and Discussion

Take BraTS-GLI-00001-000 as example. Figure 2 illustrates the visual comparison between the original native volume and the final combined structural output. A rigorous visual inspection of the synthesized pseudo-healthy structures reveals two primary remaining challenges in the current pipeline checkpoint:

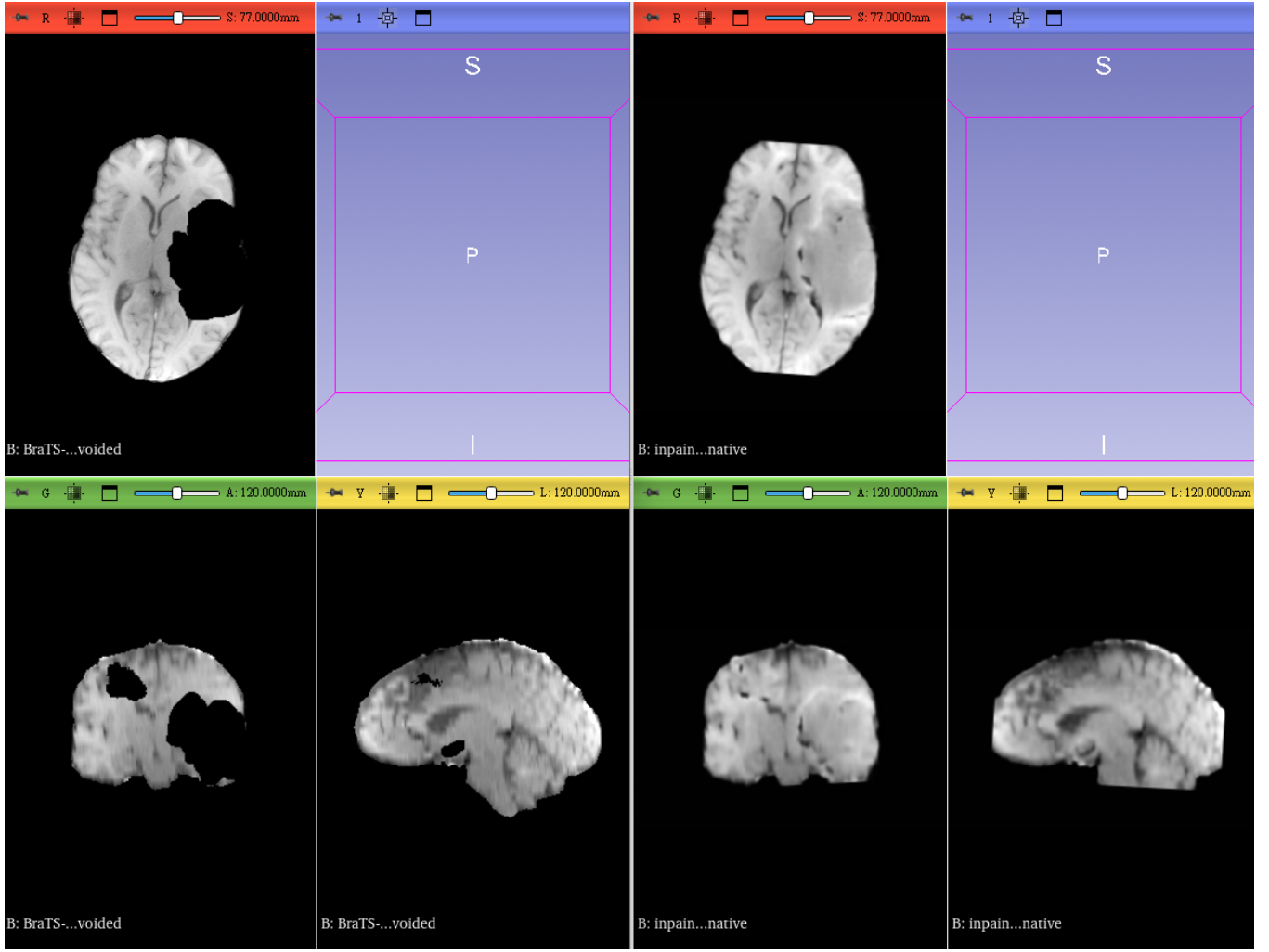


Figure 2: Qualitative inpainting results for case BraTS-GLI-00001-000.

- (1) **Boundary Discontinuity and Micro-vacuoles:** At the spatial interface transitioning from the original healthy tissue ($\mathbf{x}_{\text{voided_MNI}}$) to the freshly generated pseudo-healthy region, slight voxel-level discontinuities and microscopic structural gaps (void patches) are visible.
- (2) **Loss of High-Frequency Morphological Textures:** While the 17-channel conditioned LDM successfully reconstructs the macro-environment and global geometric symmetries of the brain, fine-grained tissue details—such as sub-voxel gray matter cortical folding patterns and deep white matter tract boundaries—appear overly regularized or partially blurred. This attenuation of high-frequency textural details highlights a minor informational bottleneck within the Stage 1 Autoencoder, wherein lossy compression into a compact 8-channel latent space inevitably discards highly intricate, patient-specific structural variations to prioritize structural stability.

Table 1: Quantitative evaluation of the proposed region-aware MNI inpainting framework on the BraTS validation set ($N = 219$). Metrics are reported as Mean $\pm$ Standard Deviation.

Metric	Value (Mean $\pm$ Std)
SSIM $\uparrow$	0.7005 $\pm$ 0.1216
PSNR (dB) $\uparrow$	16.1258 $\pm$ 2.5945
PSNR (Normalized) $\uparrow$	16.1258 $\pm$ 2.5945
MSE $\downarrow$	0.0293 $\pm$ 0.0212
RMSE $\downarrow$	0.0517 $\pm$ 0.0259
MAE $\downarrow$	0.0342 $\pm$ 0.0182
MSLE $\downarrow$	0.0101 $\pm$ 0.0095

References

- [1] Chitwan Saharia, et al. "Palette: Image-to-image diffusion models." ACM SIGGRAPH 2022 Conference Proceedings. 2022.

- [2] Andreas Lugmayr, et al. "Repaint: Inpainting using denoising diffusion probabilistic models." CVPR. 2022.
- [3] Cheng Wan, Bahram Jafraسته, Ehsan Adeli, Miaomiao Zhang, and Qingyu Zhao. "Anatomically Guided Latent Diffusion for Brain MRI Progression Modeling." *arXiv preprint arXiv:2601.14584*, 2026.